

Response to Reviewers

Comparing One-Layer and Two-Layer Economic MPC for a SAG Milling Circuit – AdCONIP 2026

We thank both reviewers for their careful reading and constructive comments. Each comment is quoted in italics, followed by our response. All changes described below are included in the revised manuscript.

Reviewer 1

Comment 1 – Typo in Eq. (2)

There is a typo in Eq. (2).

Reply. Thank you for catching this. It is a font-substitution artifact in the Word version of the manuscript. The abbreviation “s.t.” (subject to) in Eq. (2) was stored as a text run inside the equation. The latex and the PDF files are correct. We have rewritten the abbreviation as roman math letters. Reviewer 2 reported the same issue (Minor comment 2), so both reports are resolved by this fix.

Reviewer 2

Comment 1 – Symbol inconsistency for mill speed

The mill speed is labelled ω_c in Fig. 1 but φ_c in Section VII-A. Since φ_f and φ_c would then coexist with different meanings, this risks confusion. The variable should be defined once in Section II and used consistently throughout, including figures.

Reply. We checked Fig. 1 at full resolution: the mill speed is labelled ϕ_c there, and ω_c does not appear anywhere in the manuscript or its figures. The reviewer likely read a low-resolution rendering, where the small italic ϕ can be mistaken for an ω . The underlying request is implemented together with Comment 2: Section II now presents the main model equations and defines the raised variables at first use, ϕ_c (mill speed) and ϕ_f (ore hardness). The reviewer’s concern is on point, the two symbols are easy to confuse. However, we keep the ϕ notation of the underlying model literature, and all variable descriptions are included in the manuscript.

Comment 2 – Section II lacks model equations and parameter definitions

The key state equations, power-draw expression, and fines-production relation should be included. The parameters α_r and φ_f should be introduced here as part of the model description.

Reply. Agreed, and implemented. Section II now presents the main governing equations of the model, the state balances, the breakage relations and the power draw, and defines the main variables at first use, including α_r , ϕ_f and ϕ_c . The parameters α_r and ϕ_f are thereby introduced before their use in Section V.

Comment 3 – The assumption $A = \text{const}$ lacks quantitative justification

The reader needs at least an order-of-magnitude estimate of how much A varies over the PSE range used (0.65 to 0.715).

Reply. This comment is right on the spot. Historically, we used a step function for the PSE floor, with two constant values (0.65 and 0.715), to test how each control system behaves when the set-point changes. We also did not simulate the downstream circuits, flotation and leaching, which are the parts that actually compute and use $A(\text{PSE})$.

In our case, however, the choice of one or two set-points does not change the result. The study is an apples-to-apples comparison of the control designs under one common protocol. As long as the closed-loop PSE hovers close to the active set-point, $A(\text{PSE})$ can be treated as a constant to simplify the optimisation. Figure 1 shows the realised PSE of all six benchmark runs against the stepped floor: within each phase of the protocol the closed-loop PSE stays within a few hundredths of the active floor.

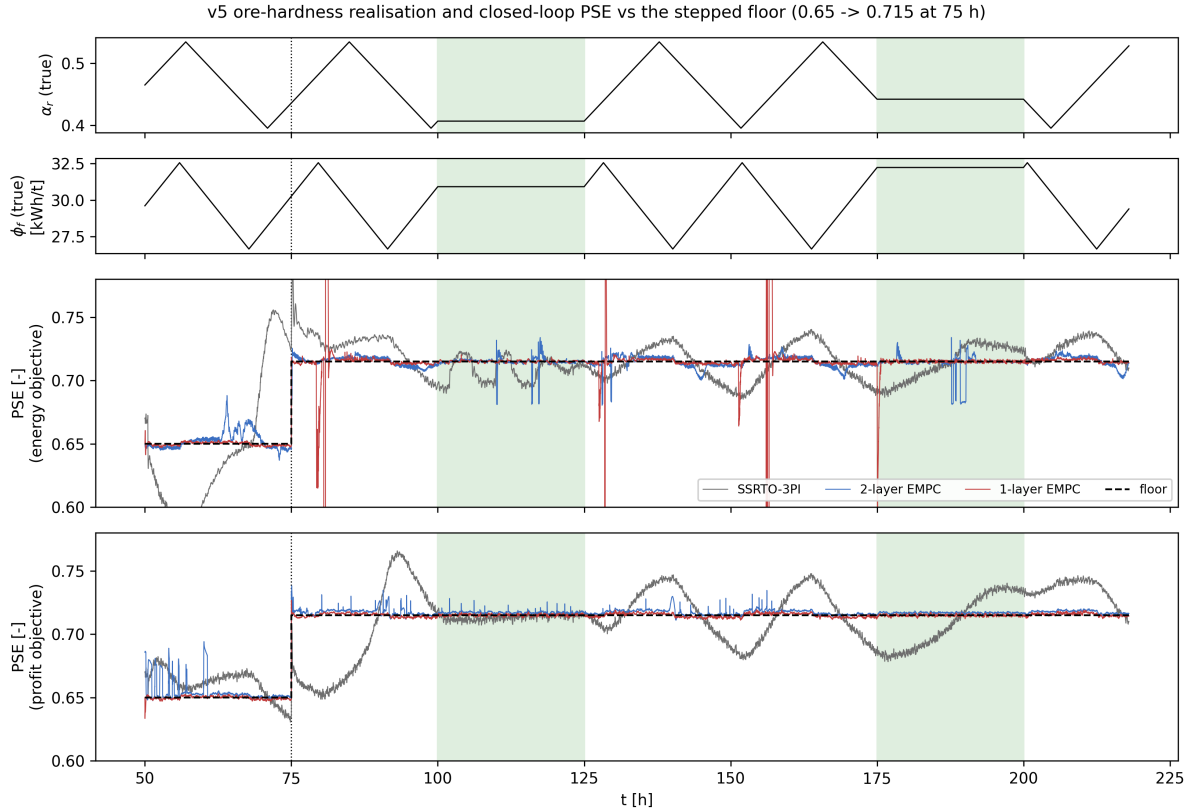


Figure 1: The v5 ore-hardness realisation (top) and the raw closed-loop PSE of the three architectures under the energy and the profit objective against the stepped product-size floor (0.65 to 0.715 at 75 h). Within each phase the PSE hovers at the active floor, which justifies treating $A(\text{PSE})$ as constant in the benchmark.

We also recomputed SEC using only the region after the 75 h step, where a single floor value is active (Table 1). The result becomes stronger: the SEC values cluster even more tightly than over the full protocol, so the single-set-point view reinforces the paper’s conclusion that specific energy is comparable across the architectures. The common upward shift of about 0.3 kWh/t reflects the finer grind the higher floor demands and affects every design equally, as Fig. 3 of the paper also shows directly.

Table 1: SEC of the six benchmark runs over the full post-warm-up window, as in Table I of the paper, and over the single-set-point region after the 75 h floor step.

Architecture	Objective	SEC, 50–218 h (kWh/t)	SEC, 75–218 h (kWh/t)
1-layer EMPC	energy	19.58	19.92
1-layer EMPC	profit	19.55	19.93
2-layer EMPC	energy	19.64	19.98
2-layer EMPC	profit	19.31	19.60
SSRTO+3PI	energy	19.65	19.94
SSRTO+3PI	profit	19.58	19.90

The reviewer’s broader point stands. If absolute economic percentages were reported for an operating plant, the stability of A over the realised PSE distribution would have to be established with the downstream circuits in the loop. That analysis is part of ongoing plant-calibration work, and in follow-up work we plan to simplify the protocol to a single set-point. Since this choice does not change our main conclusions, we keep the reported results over the full region, which keeps the tables and figures of the paper consistent.

Minor 1 – Supporting values for the order-of-magnitude claim in III-B

Section III-B asserts that revenue exceeds energy cost by more than an order of magnitude, but the supporting values ($A = 71$ \$/t, $B = 0.06$ \$/kWh) appear only in Section V. A forward reference or approximate magnitudes would help.

Reply. Agreed. The claim in Section III-B now carries a forward reference, “Revenue exceeds energy cost by more than an order of magnitude (Section V)”, pointing to Section V where the values $A = 71$ \$/t and $B = 0.06$ \$/kWh are given.

Minor 2 – “s.t.” renders as $\sigma.\tau.$ in Eq. (2)

In Eq. (2), “subject to” appears to render as $\sigma.\tau.$ rather than s.t., likely due to font substitution. Please verify.

Reply. This is also the typo reported by Reviewer 1, so both reports are resolved by this fix.

Minor 3 – Hyphenation breaks produce run-together words

Hyphenation breaks produce run-together words at several points (e.g. “orehardness,” “singlestage,” “recedinghorizon”).

Reply. We could not find any of the run-together forms in either the PDF or the Word file; all compound terms are correctly hyphenated in both. The quoted forms are likely artifacts of joining line-end hyphenated words while the paper was converted or annotated during review.